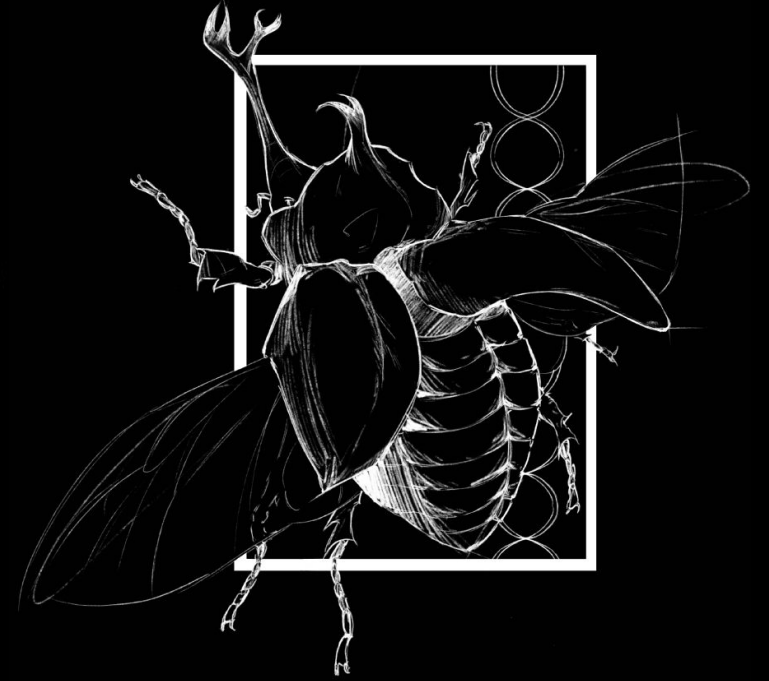




Winging It:
Environment Plays No Effect On
Wing Reduction in *Palaeognathae* (Aves)

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Background

Palaeognathae are a clade of birds commonly known for being flightless. The group includes the flightless ostriches, kiwi, cassowary, emu, and rhea, the volant tinamou, and the recently extinct giant flightless moa and elephant birds. Of these groups, all except the tinamou have reduced wing size compared to the size of their body, with ostriches and rhea having large wings with long vaned feathers, and cassowary, emu and kiwi having small reduced wings, and the moa which have no wings whatsoever. Many studies have taken a look at the loss of flight within the group, and some have examined wing usage in certain scenarios (i.e. the ostriches wing use in air braking and shielding young from the heat of the sun) but no studies have looked at why the degree of reduction is more extreme in some genera compared to others in the group. Due to this lack in our scientific understanding, this study examines the relationship between the wing length and habitat. Primary habitat and habitat density were chosen due to the relatively narrow range of environments occupied by these animals, and a suggested link between open environments and the reduction of pressure to reduce wings in response to this (i.e ostriches and rhea both live in relatively open areas and have larger proportional wings).

Study Design & AI

Methods:

Wing length was taken from AVONET, measured from the base of the carpal joint to the tip of the longest primary. Extant species primary habitat and habitat density was also taken from AVONET. Moa habitat data was inferred from the descriptions of Bunce et al., 2009, and then applied to the parameters used in the AVONET database. The average mass of extant taxa was taken from BIRDBASE, due to the availability of a more complete and accurate dataset. Extinct moa masses were taken from Latham et al., 2020. Using that data, proportional wing length was derived from dividing the ling length by the average mass. The phylogenetic tree contains all species that are currently recognized excluding *Aepyornithiformes* (which are present as genera) due to a lack of consensus. This was constructed from a combination of Yonezawa et al., 2017 for *Palaeognathae* as a whole, TToL5 for *Dinornithiformes*, and Musher et al., 2025 for *Tinamidae*. Additionally, *Struthio molybdophanes* was added using the average estimated divergence time from Freitag and Robinson, 1993 and *Casuarus unappendiculatus* was added to *Casuarus* in a polytomy, as the finer details of divergence are unknown. Following data collection, all taxa were subject to several Phylogenetic Generalized Liner Models (PGLM).

AI Usage:

Claude Sonnet 4.5 was used for streamlining and debugging code in R. All other methods, results, and writing were human generated.

Model Fitting

A Phylogenetic GLM (PGLM) with proportional wing length as the response variable and both habitat measures as predictors. PGLMs fit linear models while accounting for phylogenetic relatedness. For this model, lambda was calculated via maximum likelihood. The dataset containing all taxa were subject to three PGLMs. The first comparing wing length to primary habitat only, the second comparing wing length to habitat density only, and the third comparing wing length to both habitat traits. After discovering a lack of significance in the results, volant and flightless species were split into two groups and each subjected to additional PGLMs. The volant species were applied to an additive model with both habitat variables, while flightless species were applied to a habitat density only comparison, due to a lack of overlapping data preventing an additive model. Following the completion of each PGLM, the tests were subject to an ANOVA to determine significance and variance within each model, as well as the between the three complete dataset models.

Results

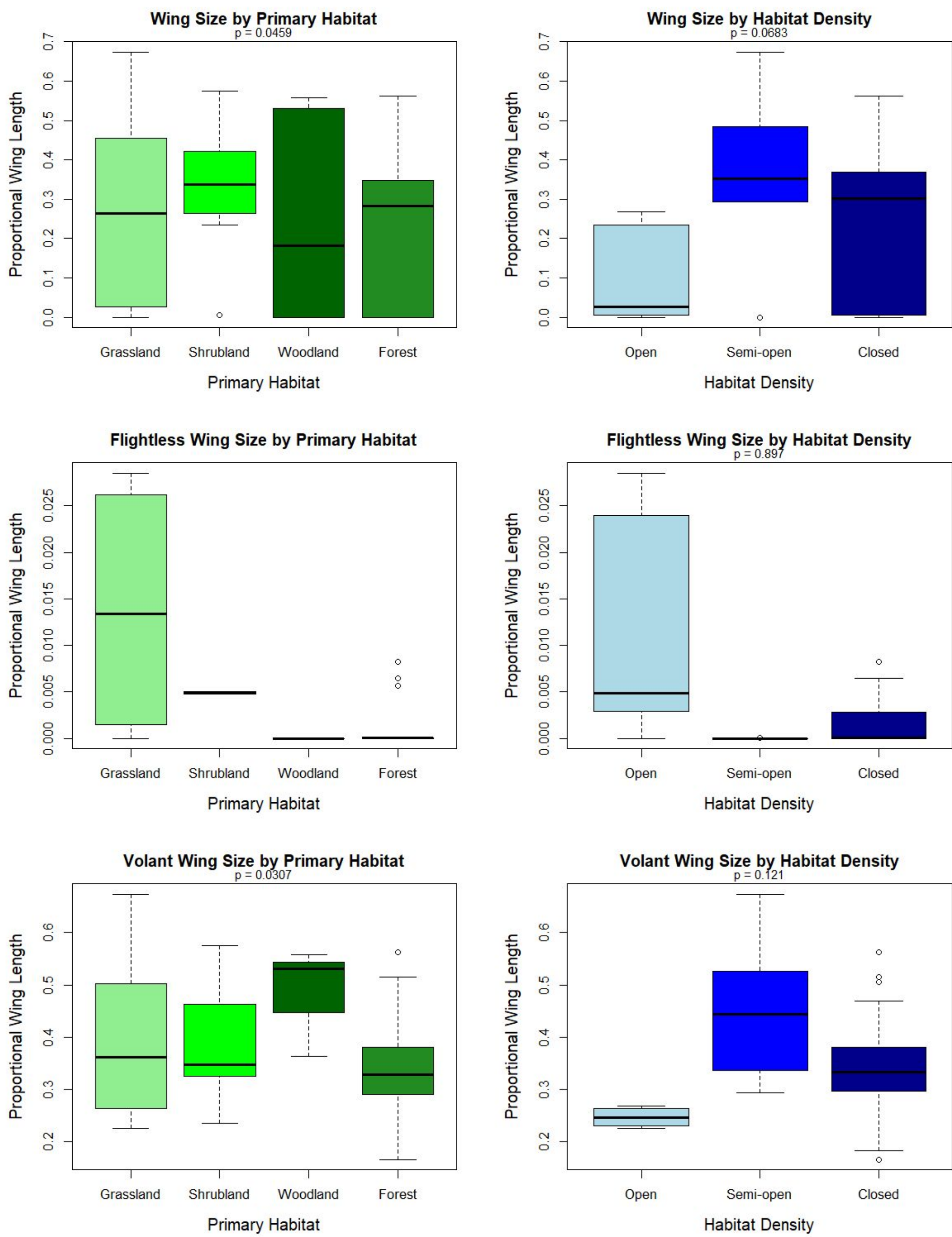
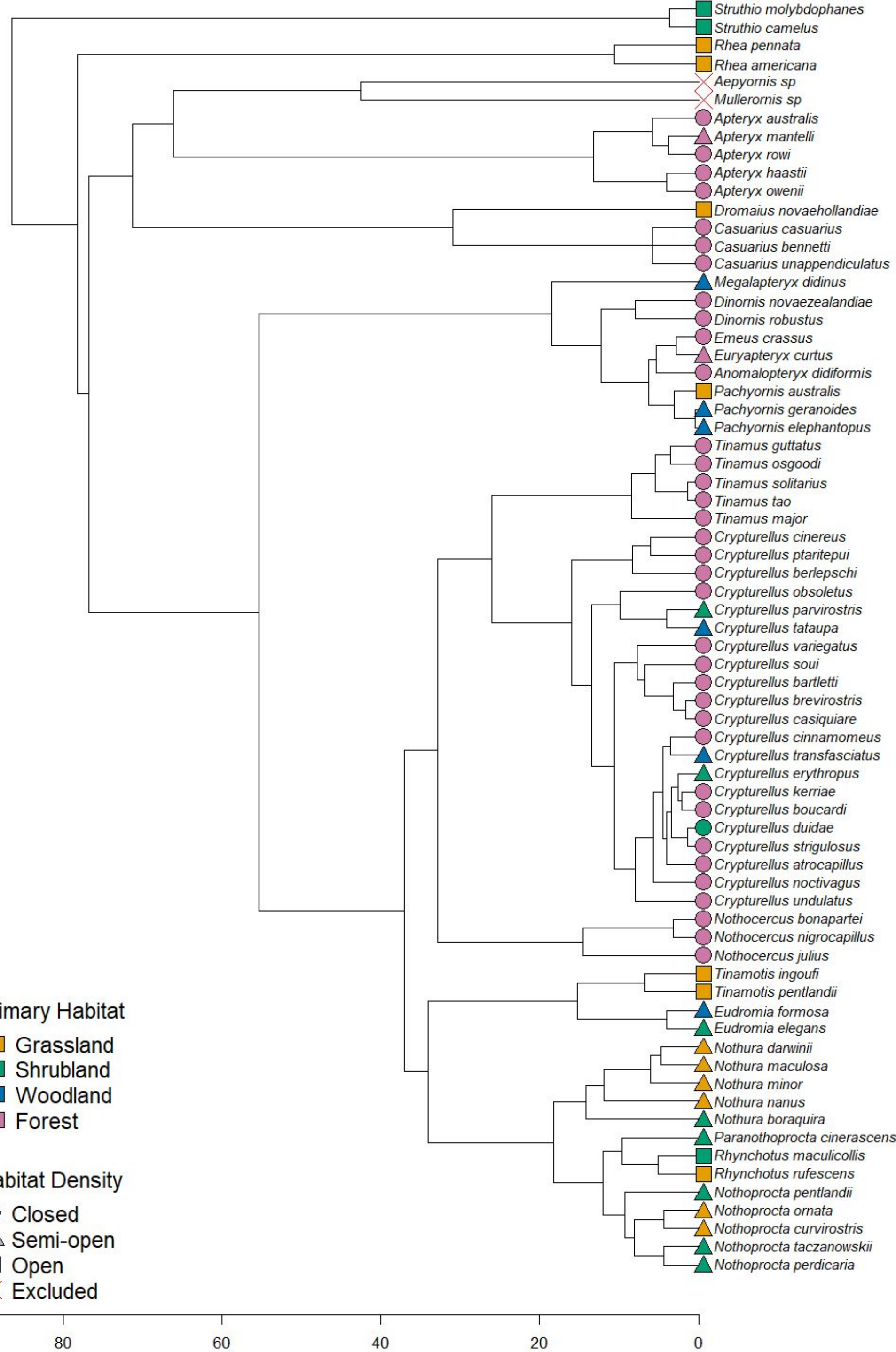


Figure 1 →
Complete species level phylogenetic tree of *Palaeognathae*. Tip labels are color coded to primary habitat, with label shape depicting habitat density.

← Figure 2
Box plots depicting the relationship of proportional wing size compared to either habitat traits individually. (Top) All taxa subjected to PGLM. (Middle) Flightless taxa only, subjected to PGLM. (Bottom) Volant taxa only, subjected to PGLM.



Discussion

Results

Comparing the results uncovered by the PGLMs, the proportional wing length within the clade as a whole has no trends relating to habitat density nor primary habitat. In all analyses *Nothura nanus*, the Dwarf Tinamou, was the only taxon with a proportional wing length that was greater than 1, with all other taxa having a value less than 0.7, causing a high degree of separation and was thus excluded in the PGLMs. Across all analyses comparing wing length to the primary habitat, no distinctive trends can be determined from the plots. The distribution of the primary habitats for the volant species and the clade as a whole suggests an even distribution, while the diversity of wing lengths in flightless species is relegated to the grasslands. Comparing the wing length to habitat density led to non-significant results across each group that was sampled.

Potential Issues

Potential explanations for the lack of significant data can be attributed to the following, with further discussion of each possibility. Modern palaeognaths are a small group with less than 70 species currently recognized (including those extinct within the past 1000 years), because of the small dataset trends are less likely to manifest compared to a larger set of species. The trait of wing length was calculated from the carpal joint to the tip of the longest primary, which artificially inflates the wing size of birds with longer feathers. Since flightless species no longer need to maintain the more rigid proportions of the underlying skeletal structure required for flight, said proportions of the wings are subjected to less constraints, and may be able to achieve “unusual” structure compared to the ancestral state. The model chosen may be unsuitable for this study as the presence of two lineages with zero or near-zero wing lengths.

Future Work

Due to the lack of a trend between the environment and wing reduction, other traits will have to be examined. Some examples of behavioral data may include the use of wings in ground locomotion, display behaviors using the wings, or use in regulating incubation temperature. Additionally, while the initial loss of flight may be related to the lack of predators in the environment, the degree of reduction is different among the genera that inhabit the island environments (Cassowary, Kiwi, Moa, and Elephant birds), which may be related to more complex environmental influence that requires reconstruction of the ancestral state.



Casuarus casuarus by John Gould (1850s)